DPP DEPARTMENTAL POLICY AND PROCEDURE

TITLE:	NEUROLOGICAL ASSESSMENT
DEPARTMENT:	INTENSIVE CARE UNIT
APPLIES TO:	ALL MEDICAL AND NURSING STAFF

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CONTENTS

1. PURPOSE	4. POLICY	7.REFERENCE/CROSS-REFERENCES
2. DEFINITION & ABBREVIATION	5. PROCEDURE	8. ATTACHMENT
3. RESPONSIBILITY	6. MEASUREMENT	9. REVISION CLAUSE

1. PURPOSE

1.1. To indicate the first sign of central nervous system deterioration.

1.2. To establish a baseline, detect subtle changes, monitor the progression of neurological conditions, and guide timely interventions to improve patient outcomes.

2. DEFINITION & ABBREVIATION

2.1. **Neurological Assessment** – monitors a critically ill patient's nervous system function and includes evaluating the level of consciousness, motor responses, pupillary function, and vital signs.

2.2. **Level of Consciousness (LOC)** – Assess mental status using tools like the AVPU scale (Alert, Verbal, Pain, Unresponsive), paying attention to any changes in attention and orientation.

2.3. **Motor Function** – Evaluate the patient's ability to move extremities, check for strength, and assess reflexes.

2.4. **Pupillary Function** – Examine pupil size, equality, and reactivity to light.

2.5. **Cranial Nerves** – Assess cranial nerve function, which is important for understanding neurological status.

2.6. **AVPU scale** – is a medical tool to quickly assess a patient's level of consciousness by categorizing their responsiveness to stimuli: Alert, responds to Voice, responds to Pain, or Unresponsive.

2.7. **Glasgow Coma Scale (GCS)** – is a neurological scale that aims to give a reliable or objective way of recording the conscious state of a person for initial as well as subsequent assessment.

2.8. **Lunula** – is the crescent-shaped whitish area of the bed of a fingernail or toenail.

3. RESPONSIBILITY

3.1. **Physician** – must undertake necessary actions and/or immediate intervention required when alteration in the level of consciousness of the patient noted.

3.2. **Head Nurse** – ensures that all staff is aware of this policy and are competency tested.

3.3. **ICU Staff Nurse** – must perform and document the neurological assessment for the patient and immediately inform the Physician on duty when alteration is noted.

4. POLICY

4.1. All Medical and Nursing Staff should adhere to this policy and procedures.

4.2. There are multiple parts to complete neurological check, typically used for Adult is the Glasgow Coma Scale.

4.3. All reversible causes must be rule out first, by identifying and treating reversible conditions that may be causing the neurological changes, such as issues with airway, breathing, circulation, glucose levels, or seizures (ABCGS).

4.4. A decreased level of consciousness or a new neurological change must be reported promptly to ensure timely medical intervention.

4.5. All ICU staff must be competent in performing the neurological examination.

5. PROCEDURE

5.1. Identify the patient correctly using two identifiers. (Refer to APP-QPS-0016 Patient Identification).

5.2. Review the patient's neurological assessment baseline as available. Always compare current findings to the patient's baseline neurological status.

5.3. Take into your account the present condition of the patient which can affect the neurological assessment such as sedation, or Intubation.

5.4. **Mental Status & Level of Consciousness**:

5.4.1. Assess the patient's responsiveness and orientation to person, place, and time:

5.4.1.1. **Person:** The individual's awareness of their own name, age, and personal attributes.

5.4.1.2. **Place:** Their understanding of where they are, such as the type of facility (hospital, ICU) or the city they are in.

5.4.1.3. **Time:** Knowledge of the current date, time of day, and potentially the day of the week.

5.4.2. Determine the level of consciousness using Glasgow Coma Scale (GCS). With grades and starting from most severe.

5.4.2.1. **Eye Response (E)**

5.4.2.1.1. (1) – No eye opening

5.4.2.1.2. (2) – Eye opening in response to pain stimulus. (A peripheral pain stimulus such as squeezing the Lunula area is more effective)

5.4.2.1.3. (3) – Eye Opening in response to voice

5.4.2.1.4. (4) – Eye opening spontaneously

5.4.2.2. **Verbal Response (V)**

5.4.2.2.1. (1) – No verbal response

5.4.2.2.2. (2) – Incomprehensible sounds

5.4.2.2.3. (3) – Inappropriate /Incoherent

5.4.2.2.4. (4) – Confused/Disoriented

5.4.2.2.5. (5) – Oriented, Converses normally

5.4.2.3. **Motor Response (M)**

5.4.2.3.1. (1) – No movement

5.4.2.3.2. (2) – Decelerate Response (Extension to painful stimuli)

5.4.2.3.3. (3) – Decorticate Response (Abnormal flexion to painful stimuli)

5.4.2.3.4. (4) – Withdrawal to painful stimuli (Flexion)

5.4.2.3.5. (5) – Localizes to painful stimuli

5.4.2.3.6. (6) – Obeys Command

5.4.3. Determine the level of consciousness using the AVPU scale.

5.4.3.1. **Alert:** Check if the patient is awake and aware of you.

5.4.3.2. **Voice**: If not alert, speak loudly to the patient to see if they respond.

5.4.3.3. **Pain**: If no verbal response, apply a painful stimulus as a sternal rub or pinching the earlobe to check for any reaction.

5.4.3.4. **Unresponsive**: If there is no response to any of the above, the patient is unresponsive.

5.5. **Cranial Nerve Assessment:**

5.5.1. Perform Pupil check:

5.5.1.1. Darken the room or let the patient close eyes for 30 seconds prior to testing.

5.5.1.2. Using a penlight or flashlight check each pupil separately and then at the same time for size and shape, equality of reaction and consensual reaction. The light used for checking must be brighter than the room light.

5.5.2. Check extra ocular eye movement. Ask the patient to have eyes follow an object without moving the head.

5.5.3. Check facial symmetry by asking the patient to smile or wrinkle forehead. Observe for symmetrical forehead, lip and facial muscle movement.

5.5.4. Assess speech for appropriateness, clarity, expressiveness and receptive components

5.5.5. Assess the Swallowing, note difficulty swallowing or changes in speech.

5.5.6. Assess cough Reflex in intubated patients by observing a patient's response to airway stimulation, such as tracheal suctioning.

5.6. **Motor & Sensory Function:**

5.6.1. Check motor function by asking the patient to follow the commands:

5.6.1.1. Grip onto and release your hands

5.6.1.2. Hands their arms against you.

5.6.1.3. Pull their arms away from you.

5.6.1.4. Extend and flex both feet against resistance (i.e. your hands)

5.6.1.5. Raise both legs equally off bed.

5.6.1.6. Hold both arms up with palms toward ceiling with eyes closed for 10 seconds (observe any turning of the hands or if the arms fall or drift)

5.6.2. Determine response to physical stimulus if unable to follow commands, place arms away

from patient's side or abdomen to determine if patient moves arm up or down.

5.6.3. **Rate extremity strength** as follows:

5.6.3.1. (5) – strong against resistance

5.6.3.2. (4) – breaks against resistance

5.6.3.3. (3) – moves against gravity

5.6.3.4. (2) – unable to move against gravity

5.6.3.5. (1) – flicker

5.6.3.6. (0) – no response

5.6.4. Check sensory function by observing patient's response to touch if conscious. Ask the patient about any numbness/tingling or pain in face and extremities.

5.7. **Obtain vital signs**

5.8. **Coordination and Balance:**

5.8.1. Gait: If the patient can ambulate, assess their ability to walk with steady balance.

5.8.2. Truncal Stability: Observe for stability of the trunk, particularly if the patient has had issues with balance or falls.

5.8.3. Determine need for implementing Fall Prevention Protocol.

5.9. Score the Glasgow Coma Scale of the patient based on assessment of the physician.

5.10. Perform the neurological assessments every four hours or more frequently, depending on the patient's condition.

5.11. Document the assessment in Nurses progress notes in the MCC and ICU Reassessment Form.

6. MEASUREMENT

6.1. 100% of the medical and nursing staff are well oriented and adheres to this policy.

7. REFERENCES /CROSS- REFERENCES

7.1. https://en.wikipedia.org/wiki/Glasgow_Coma_Scale

7.2. Cbahi.org

7.3. Mills, E. (2004). Nursing Procedures,4th Ed. Philadelphia. Lippincott Williams & Wilkins.

7.4. http://en.wikipedia.org/wiki/Glasgow_Coma_Scale

8. ATTACHMENT

8.1. Neurological Assessment Competency Check Off

9. REVISION CLAUSE

9.1. Changes to the hospital policy and procedure template based from the Al-Baha Health Cluster logo,

font face and design

9.2. Modification of definition and abbreviation, responsibilities and procedures.